5. $SE = \sqrt{\frac{36}{81} + \frac{25}{64}} = .914$ Standard error

$Z = \frac{14-12}{.914} = 2.19$ test statistic

6. $SE = \sqrt{\frac{2.25}{49} + \frac{1.96}{64}} = .277$

$Z = \frac{3.2 - 2.8}{.277} = 1.45$

95% CI $.4 \pm 1.96(.277) = (-.14, .94)$

We fail to reject the null hypothesis. 0 is inside the confidence interval